Bank Customer Churn Prediction

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1. Data Understanding **3**
1.1 Dataset Overview **3**
1.2 Data Quality Assessment **3**
✔ Missing Values 3
✔ Duplicate Records 3
✔ Data Type Validation 4
1.3 Variable Description **4**
1.4 Target Variable Definition **6**
Churn 6
Overall churn rate: ≈ 20% 6
1.5 General Dataset Characteristics **6**
1.6 Analytical Suitability **6**
1.7 Summary **7**
1. Data Preparation and Cleaning **8**
1.1 Missing Value Assessment 8
1.2 Duplicate Record Check 8
1.3 Data Type Validation 9
2. Feature Engineering (Helper Columns) **10**
2.1 Age Group Classification 10
2.2 Balance Category Classification 10
3. Key Performance Indicators (KPIs) **11**
3.1 Total Customers 11
3.2 Churn Rate 11
3.3 Average Balance 11
3.4 Average Credit Score 12
4. Exploratory Data Analysis (Pivot Tables) **12**
4.1 Churn by Gender 12
Finding: 12
4.2 Churn by Age Group 12
Finding: 12
4.3 Churn by Balance Category 13
Findings: 13
4.4 Churn by Active Membership 13
Finding: 13
Executive Summary: Geographic and Demographic Churn Drivers **13**
5. Business Insights **14**
5.1 High-Risk Segment: Age 46–55 14
5.2 Low-Balance Customers 15
6. Strategic Business Recommendations **15**
6.1 Loyalty Program for 40+ Customers 15
6.2 Increase Engagement of Inactive Members 16
6.3 Incentivize Low-Balance Customers 16
6.4 Cross-Selling Strategy 17
7. Analytical Workflow (Professional Perspective) **17**

1. Data Understanding

Total Records	10,000 customers
Total Variables	12 columns
Dataset Shape	(10000, 12)
Missing Values	None detected
Duplicate Records	None found
Data Formatting	Correctly structured

1.1 Dataset Overview

1.2 Data Quality Assessment

A full data validation process was conducted to ensure analytical reliability.

✔ Missing Values

- No blank or null entries identified.

✔ Duplicate Records

- `customer_id` verified as unique.

- Each row represents a distinct customer.

✔ **Data Type Validation**

- Numeric variables properly formatted.
- Binary variables correctly encoded (0 = No, 1 = Yes).
- Categorical fields standardized (e.g., Male/Female, Country names).

No structural or formatting issues were identified.

1.3 Variable Description

No	Variable	Description	Type
1	customer_id	Unique identifier for each customer	Numeric (Identifier)
2	credit_score	Customer creditworthiness score	Numeric
3	country	Country of residence	Categorical
4	gender	Customer gender	Categorical
5	age	Customer age in years	Numeric

6	tenure	Number of years with the bank	Numeric
7	balance	Account balance	Numeric (Currency)
8	products_number	Number of bank products used	Numeric
9	credit_card	Credit card ownership (1=Yes, 0=No)	Binary
10	active_member	Account activity status (1=Active, 0=Inactive)	Binary
11	estimated_salary	Estimated annual salary	Numeric
12	churn	Customer exit status (1=Left, 0=Stayed)	Binary (Target Variable)

1.4 Target Variable Definition

Churn

Value	Meaning
1	Customer left the bank
0	Customer remained with the bank

Overall churn rate: **≈ 20%**

This variable represents customer attrition and serves as the primary outcome for retention analysis.

1.5 General Dataset Characteristics

- Clean and analysis-ready structure
- Balanced numeric and categorical features
- No data integrity issues
- Suitable for segmentation and behavioral analysis

1.6 Analytical Suitability

The dataset supports the following analytical techniques:

- Pivot table analysis
- KPI computation (churn rate, averages, segmentation metrics)
- Customer segmentation
- Risk profiling
- Dashboard development
- Business intelligence reporting

1.7 Summary

The dataset demonstrates high data quality with no missing or duplicate records. All variables are properly formatted and logically structured.

Given its completeness and clarity, the dataset is fully suitable for conducting customer churn analysis and generating strategic business insights.

Next

1. Data Preparation and Cleaning

1.1 Missing Value Assessment

A full dataset review was conducted to identify any missing or blank values.

Method used:

- Excel: `Ctrl + G → Special → Blanks`
- Column-level filtering for verification

Result:

- No missing values detected.
- Dataset is complete and suitable for analysis.

1.2 Duplicate Record Check

To ensure data integrity, duplicate records were examined.

Method used:

- Conditional Formatting → Highlight Duplicate Values
- Focused on `customer_id` column

Result:

- No duplicate `customer_id` values found.
- Each row represents a unique customer.

1.3 Data Type Validation

Proper data types were verified to avoid analytical errors.

Column	Expected Type	Status
credit_score	Numeric	Valid
age	Numeric	Valid
balance	Numeric (Currency)	Valid
estimated_salary	Numeric	Valid
churn	Binary (0/1)	Valid

Validation Methods Used

Visual Check:

- Numeric values aligned to the right.
- Text values aligned to the left.

Formula Validation (Professional Check):

=ISNUMBER(G2)

Result:

- TRUE → Correct numeric format

- FALSE → Stored as text (converted where necessary)

Dataset confirmed properly formatted.

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2. Feature Engineering (Helper Columns)

To enhance analytical capability, additional business-oriented variables were created.

2.1 Age Group Classification

Created new column: `age_group`

Purpose:

- Enable segmented churn analysis.

Formula used:

=IF(E2<=25,"18-25",

IF(E2<=35,"26-35",

IF(E2<=45,"36-45",

IF(E2<=55,"46-55","56+"))))

2.2 Balance Category Classification

Created new column: `balance_category`

Purpose:

- Analyze churn behavior by financial segment.

Formula used:

=IF(G2=0,"Zero Balance",

IF(G2<50000,"Low",

IF(G2<150000,"Medium","High")))

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3. Key Performance Indicators (KPIs)

3.1 Total Customers

=COUNTA(A2:A10001)

Total customers: **10,000**

3.2 Churn Rate

Churn defined as:

- 1 = Customer left
- 0 = Customer stayed

Formula:

=AVERAGE(L2:L10001)

Churn Rate:
20.37%

Verification:

=COUNTIF(L2:L10001,1)

Customers churned: 2,037

3.3 Average Balance

=AVERAGE(G2:G10001)

3.4 Average Credit Score

=AVERAGE(B2:B10001)

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4. Exploratory Data Analysis (Pivot Tables)

4.1 Churn by Gender

Pivot Configuration:

- Rows → Gender
- Values → Average of churn

Finding:

- Female customers exhibit higher churn rate than male customers.

4.2 Churn by Age Group

Pivot Configuration:

- Rows → age_group
- Values → Average of churn

Finding:

- Highest churn observed in **46–55 age group**.

4.3 Churn by Balance Category

Pivot Configuration:

- Rows → balance_category
- Values → Average of churn

Findings:

- Low balance customers have the highest churn.
- Zero balance customers are not the highest churn group.

4.4 Churn by Active Membership

Pivot Configuration:

- Rows → active_member
- Values → Average of churn

Finding:

- Active members churn significantly less.
- Engagement reduces churn probability.

Executive Summary: Geographic and Demographic Churn Drivers

1. The Geographic Variance (The "Where")

While the global churn rate is **20.37%**, this figure is heavily influenced by the large customer base in **France** (50% of the dataset). However, **Germany** is the highest-risk market with a **32% churn rate**—double that of France (16%) and Spain (17%).

- **Actionable Insight:** The business must investigate if the German market has a more competitive banking landscape or if there is a specific service dissatisfaction unique to that region.

2. The Age Demographics (The "Who")

Churn is not uniform across age groups. In all three countries, customers in the **46–55** and **36–45** age brackets show the highest volume of exits.

- **Actionable Insight:** Younger customers (18–25) are remarkably loyal, with very low churn counts. The bank's retention strategy should focus on "Mid-Life" customers, perhaps through improved long-term wealth management or loyalty incentives tied to age-specific financial goals.

3. Balance-Driven Behavior (The "Financial Profile")

The "Balance Category" dashboard reveals a critical behavior:

- In **France**, customers with a **Low Balance** are at extreme risk, with a **49% churn rate**.
- In **Germany**, the risk shifts toward **Medium Balance** holders (35% churn).
- **Actionable Insight:** This suggests that "Low Balance" French customers may be leaving because of high account fees, while "Medium Balance" German customers might be moving their funds to competitors offering better interest rates.

5. Business Insights

5.1 High-Risk Segment: Age 46–55

Possible explanations:

- Transition toward retirement planning

- Switching to investment-focused banks

- Financial maturity leading to comparison shopping
- Account consolidation behavior

Balance alone does not fully explain churn in this segment.

5.2 Low-Balance Customers

Low balance customers are more likely to churn because:

- Lower switching cost
- Lower emotional and financial attachment
- Limited product engagement
- Reduced perceived value

This behavior aligns with common banking retention patterns.

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6. Strategic Business Recommendations

6.1 Loyalty Program for 40+ Customers

Implementation:

- Identify age_group 46–55
- Prioritize inactive and low-balance customers
- Offer:
    - Retirement planning services
    - Preferential interest rates

    - Dedicated relationship management

Measure impact:

- Compare churn rate before and after program implementation

6.2 Increase Engagement of Inactive Members

Target:

- active_member = 0

Actions:

- Mobile banking campaigns
- Fee incentives
- Customer education programs

Measure:

- Activation rate
- Churn reduction post-activation

6.3 Incentivize Low-Balance Customers

Actions:

- Bonus interest for maintaining minimum balance
- Cashback programs
- Product bundling incentives

Measure:

- Balance growth
- Reduction in churn rate

6.4 Cross-Selling Strategy

Analyze:

- Churn by `products_number`

Typical pattern:

- More products → Lower churn

Strategy:

- Offer additional services to single-product customers
- Increase switching cost via engagement depth

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7. Analytical Workflow (Professional Perspective)

This project follows a real-world banking analytics cycle:

1. Data cleaning and validation
2. Exploratory segmentation analysis
3. Identification of high-risk segments
4. Strategic recommendation design

5. Impact measurement planning

This reflects practical business intelligence methodology used in financial institutions.